

BTI 3021: Networking Project - Sprint 3

Christian Grothoff

1 Introduction

For this sprint you will write an IP router, building upon the results from your sprint.

While the driver and skeleton you are given is written in C, you may again use *any* language of your choice for the implementation (as long as you extend the `Makefile` with adequate build rules). However, if you choose a different language, be prepared to write additional boilerplate yourselves.

How an IP router works is expected to be understood from the networking class. If not, you can find plenty of documentation and specifications on the Internet.

The basic setup is the same as in the first two sprints.

1.1 Deliverables

There will be three main deliverables for the sprint:

Documentation You must document the main data structure (ARP table) and a **comprehensive test plan**.

Implementation You must implement the ARP protocol. Your implementation must answer to ARP requests, and also itself have the ability to issue ARP requests and to cache ARP replies. For this, you are to extend the `arp.c` template provided (or write the entire logic from scratch in another language).

Testing You must implement and submit your own **test cases** by *pretending* to be the network driver (see below) and sending ARP requests or command-line inputs to your program and verifying that it outputs the correct frames. Additionally, you should perform and **document interoperability** tests against existing implementations (i.e. other notebooks from your team to ensure that your ARP protocol implementation integrates correctly with other implementations).

All deliverables must be submitted to Git (master branch) by **January 22nd 2021, 18:00 CET**.

1.2 Team

You are expected to work in a team of four students. The suggested division of work is:

1 Student Scrum master, also responsible for the documentation

1 Student Implementation

2 Students Testing

1.3 Grading

You get points for each of the key deliverables:

Technical documentation (A&D, test plan)	3
Correct implementation	9
Comprehensive test cases	4
Quality of English in documentation	3
Total	19

We will specifically also look for the following properties of a router:

- Correct forwarding?
- Correct address resolution and caching?
- Correct IP handling?
- Correct IP fragmentation?

Partial points are awarded.

You can earn 16, 15 and 19 points in the three networking sprints, for a total of 50 points. You need **37 points** to pass the networking component of BTI 3021.

2 Detailed interface specification

Implement **router** which routes IPv4 packets:

1. Populate your routing table from the network interface configuration given on the command-line using the same syntax as with the **arp** program.
2. Use the ARP logic to resolve the target MAC address. You **MUST** simply drop IP packets for destinations where you do not yet have the next hop's MAC address, but you **MUST** then issue the ARP request to obtain the destination's MAC instead (once per dropped IP packet).
3. Make sure to decrement the TTL field and recompute the CRC.

4. Generate ICMP messages for “no route to host” and “TTL exceeded”.
5. Support the syntax `IFC[R0]=MTU` where `MTU` is the MTU for IFC. Example: `eth0=1500`. Implement and test IPv4 fragmentation (including *do not fragment*-flag support).
6. Support dynamic updates to the routing table via `stdin`. Base your commands on the `ip route` tool. For example, “route list” should output the routing table, and “route add 1.2.0.0/16 via 192.168.0.1 dev eth0” should add a route to 1.2.0.0/16 via the next hop 192.168.0.1 which should be reachable via `eth0`. Implement at least the `route list`, `route add` and `route del` commands.